

Arrows of Time for Large Language Models

Vassilis Papadopoulos Jeremie Wenger Clement Hongler

CSFT/Institute of Mathematics, EPFL, Lausanne, Switzerland

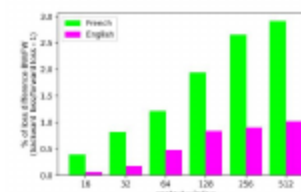
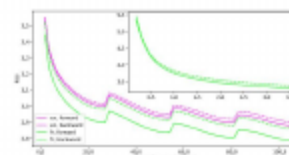
FSL/Institute of Physics, EPFL

Department of Computing, Goldsmiths, University of London, London, UK

May 24, 2026

We show a robust Arrow of Time in LLMs

- We show that backward (previous-token) models have consistently higher validation loss than forward (next-token) models across languages.
- Example (GPT2-M, $n=256$): English +0.76%, French +2.65% BW over FW; the gap appears early and persists during training.
- We find the AoT strengthens with longer context (16→512), implicating long-range dependencies.



Navigation icons

Navigation icons

How we tested it (symmetric FW vs BW)

- We train matched forward and backward autoregressive models with identical data splits, hyperparameters, architectures, and budgets.
- We evaluate cross-entropy/log-perplexity on CC-100 languages with context windows 16–512 and model sizes from 5M to 1.6B.
- We control for tokenizer direction and initialization; the FW advantage remains.

Navigation icons

Why it matters and our intuition

- We hypothesize the AoT arises from computational complexity and sparsity: forward mappings in natural language are sparser/easier to represent than their inverses.
- This explains why AoT grows with context and capacity, despite information-theoretic reversibility.
- Implications: we favor forward pretraining, design evaluations that stress long-range dependencies, and treat AoT as a diagnostic of structured data.

Navigation icons